WasteSort

Preparing the dataset

Day 2 · Tuesday



Goal Diggers
BCS34

What the inventory records

One row represents one source image.

Field	Example
Path	data/raw/plastic/plastic5.jpg
Material label / code	plastic / 4
Format / color	JPEG / RGB
Width × height	512 × 384 pixels

Included/excluded status and split are recorded separately.

Class counts before and after cleaning

Material	Original	Usable	Excluded
Cardboard	403	403	0
Glass	501	498	3
Metal	410	409	1
Paper	594	594	0
Plastic	482	480	2
Trash	137	137	0
Total	2,527	2,521	6

Paper is the largest class. Trash is the smallest.

Duplicate checks prevent misleading evaluation



glass/glass115.jpg



metal/metal91.jpg

Identical pixels, conflicting labels: both excluded.

Six excluded files. 45 near-duplicate candidate pairs for review.

Separate training, validation and test images

Partition	Images	Purpose
Training	1,774	Fit the model
Validation	371	Select settings and model
Test	376	Evaluate the frozen choice

Seed 42. Candidate groups stay in one partition.

2,521 usable images; approximately 70 / 15 / 15.

Labels become category identifiers

Material	Code
cardboard	0
glass	1
metal	2
paper	3
plastic	4
trash	5

The numbers identify classes.

They do not measure quantity or rank materials.

Use the same mapping throughout the project.

Preprocessing depends on the model

Model family	Image input	Transformation
SVM / Random Forest	64 × 64 grayscale	1,764 HOG features
Small CNN	160 × 160 RGB	Pixels scaled to [0, 1]
MobileNetV2	160 × 160 RGB	Pretrained normalization

Random augmentation applies only to training images.

Square resizing standardizes shape but changes aspect ratio.

Tuesday handoff

Inventory	All 2,527 source images and their status
Training manifest	2,521 usable images with fixed partitions
Preprocessing notebook	Encoding, resizing and input-batch checks
Human review worksheet	Labels and candidate duplicates to inspect

Ready for the next training activity.